ActInf Textbook Group ~ Cohort 5 ~ Session 2 (Chapter 1, Part 1) ~ 9/26/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 2 (Chapter 1, Part 1) ~ 9/26/2023 | 1:03:33

all right hello everyone it is
cohort five meeting two and we're in our
first discussion of chapter one so to
begin with
what is anyone's just First Take or on
chapter one or where does anybody want
to begin
and then we'll look at the specific
questions but is there any just first
overviews or just
pieces that people want to share about
chapter one or just this first
discussion
you can just go for it if you want to
ask me
so my biggest um
question is around surprise and what the
I guess the parameters and boundaries of
that is
well
sorry I think uh a part of my long time
persisting uh
the literature is like the use of some
terms like these and these might be
because I'm also a non-native speaker
but like the term of like a normative
principle like
still not sure what it means or like you
know later on they say like epistemic
this and stuff and I'm just like very
confused by those uh terms uh so maybe
clarifying yeah
what it means to be a normative
principle will be a very useful thing
for me

okay let's just see if it's been asked
overall
because as Ollie can probably attest to
let's just like at least think of oh
does anyone want to add a thought on
this like just on uses of terms overall
or we can look more specifically at what
is meant by normativity like in chapter
one
foreign
was also
having some
um uncertainty about the use of the term
I had pre-released the I had an
understanding that normative
uh
the term described something that uh the
way that some something should be
as opposed to
being descript a descriptive
theory for example which is which is
simply
stating what is observed but I
uh in the way that it's introduced here
normative is used more
I think
uh placing an emphasis on the ability to
to look at a quantity
and measure some behavior in terms of
that quantity and
um
uh emphasizing that's better Behavior or
behavior that's more in alignment with
with the theory
is suggesting will will optimize that
quantity
uh that's my understanding so far
but I'd like to

be more clear
thank you Ali go for it
yeah so normative
um I mean the use of the the term
normative uh here in active inference
literature is closely related to the
meaning of this term in uh say
philosophical literature and stuff uh so
uh basically in uh philosophy or more
specifically in ethics when we talk
about the normative Behavior or
um I don't know a normative
action uh it's uh it's interpreted as
something ethical uh or moral and the
reason we ascribe this kind of meaning
to normal to this term normative is
because we have a standard of action or
behavior against which we can compare
our own action and then judge by way of
comparing it whether it is ethical or
not so by the same token we can talk
about normativity here in this context
of active inference when we talk about a
standard for optimization of our action
and perception
and namely the free energy against which
uh the uh the error or the deviation of
the action and perception from that
standard can be evaluated and then
corrected or accounted for so that's why
uh I mean researchers in active
inference area use the term normal
normative or normativity to refer to
this kind of
compare comparison between the
comparison between the action and a kind
of a standard for optimization of that
action

I would kind of question whether that
would be tied to ethics though because
Norms don't necessarily
align with ethical behaviors and of
course ethical is subjective but you
know
um Norms have been you know for example
been leaning more toward coercive
and exploitive behaviors which in my in
my world is not ethical but uh
you know say I don't let I'd like to see
the
you know those two tied apart especially
as far as how people act because ethics
are you know are not
are not Universal
um and coerced coercive behaviors
uh tend to you know get more
get more benefits in U.S society but
maybe not other places
yeah I guess it's not like so used to me
the connection with that it's you know
it comes to these ideas from like uh
searching first principle Voice or
understands uh self-organization of main
components
so my feeling is that
you know there's nothing about our
subjective ethical more or more things
here so some so yeah I mean the fact
that we all have this uh confusion
really reveals that maybe normally
shouldn't be the work that this thing is
here because it's just like well we have
so many meanings in so many contexts so
I don't know
it is very cultural
but perhaps I mean if we look at it as

just a standard for optimization within
a certain a certain definition of a
system then
you know if we don't get too caught up
on our whatever our conceived Notions or
subjective Notions of normative as in
normal and social norms but just look at
it as a optimal Behavior within the
parameters of the system
hmm
yeah so um if I may um clear that up a
little bit well uh
I mean what I meant it was not that uh
uh the the word normativity is used in
exactly uh the same way as it's used in
I don't know philosophical literature or
ethics uh it's just
um I mean a related term that's used in
its technical sense in active inference
literature so I agree that we can maybe
uh it's too naive to take that same
meaning from philosophical literature
and use it directly in this context uh
to refer to something I don't know uh as
a social Norm or a standard or whatever
but uh again in this technical context
um Norm here just means a standard
against which the optimization of the
system happens right so uh that's what
I'll wear there is to it so uh it's not
something uh
it's not something that can be applied
to derive any kind of Ethics or morality
or whatever uh it's just a way of
describing a system uh against a
specific standard for optimization
namely uh the free energy so that's why
this term is used here and also

if we want to uh use the first principle
way of describing things we need to have
these kinds of uh
precise uh precise definitions for those
standards for optimization uh because uh
it's a way of rigorously building up uh
our descriptive model so that's
necessary to have a very strict and
rigorous
definition for any standard if we want
to build up models rigorously enough to
be useful for modeling any complex
behavior of
um cognitive agents
so basically if I understand from this
discussion can I you know when in part
1.4.1 says active inference is a
normative framework to characterize base
optimal behavior and cognition in living
organisms
does that mean that what it's saying is
that
active inference as a theoretical
framework just sets the objective
function of what systems must do which
is to minimize the original free energy
so normative just refers to
the fact that active influence
establishes the rules by which systems
have to play kind of thing is is that a
good interpretation of what normative
means yeah more or less that's what all
there is to it yeah and also there's a
more explicit definition of this term
normativity in I guess
notes one from chapter one and Page uh
let me see uh
so yeah here the term normative means

there is some evaluative standard
against which behavior can be scored
active inference is normative in the
sense that perception and action are
scored by free energy a quantity we will
unpack throughout this in the next
chapters
these are all really great points one
analogy that I find very useful or a set
like more machine learning and more
classically statistical
is by now the variational free energy
playing the role of the L2 regression
fitting Norm in linear modeling so
linear modeling doesn't just mean
straight lines so check out the earlier
friston fmri work on SPM which extends
generalized linear
modeling so fitting the L2 Norm doesn't
mean that the model is adequate it
doesn't mean it's going to work if
something changes about the underlying
data process and so on like it's none of
no modeling process can say anything
about how well the outcome
works
but this loss function the L2 norm and
linear regression or like the loss
function in uh training a neural network
these are analogous to the variational
free energy such that however you set up
a generative model it's going to have
this like base optimal
like it it's it's going to look like
it's on its path of least action given
how it's set up
with this loss function so I agree with
all the absolutely it's very it's a more

of like a methodological or a
computational normativity
in terms of being variational free
energy being more like a loss function
okay that's one that's one angle on the
normativity but I think this is also one
discussion that we returned to many
times which is just
how does something that can describe
anything like we describe things to the
ex to the extent of their thingness for
some statistical pattern
or for all these other ways of modeling
which is kind of what this
um approach ends up looking like in
generating these cognitive models like
how how can it be measuring everything
what is what is something that measures
everything say what is the uh
meter stick say
so understanding like measurement and
then all this season
so so that's that's an interesting um
um concept so it's it's not pointing
toward
um the outcome as much as it is the
value
the optimized value would that be in
other words
you know or is the model
you know I understand it's ideology free
um or you know I'm assuming I'm I'm
inferring that
um but you know you the
the perception is
uh aiming toward a goal so the goal
being a value
versus an outcome

there's a lot there but it it there's
just and of course these are just like
very various different people's
perspective on it but in one view it has
equal ideological status to a
measurements tool as a measurement tool
the kind of narrow sense which it
actually looks like when we get through
chapter six the recipe for making the
active inference generative model which
was like alluded to earlier like yeah
that's the difference between having a
generated model once the generative
model is specified slash to the extent
it's specified
it is going to do the most statistically
likely thing
but that can include a noisy
distribution
for perception action
so something being uncertain doesn't
mean that it's not the likeliest path
that is done in the model with like a
parameter that describes how noisy
something should be
that's like variance estimation Bayesian
hyper parameters
so just because it seems it it shouldn't
feel constraining in terms of
um developing like a phenomena that
seeks information or something like this
normativity isn't a
like far-reaching claim
and then there then there's all of these
other kind of
aspects that come into playing the
actual modeling
mostly from the system of Interest

just like people use linear modeling
across fields
and then there's statistics into all of
the development that happens in
statistics
just that that's one kind of
deflationary view
that is more like the statistical
parametric mapping earlier for instant
work
um and also one other point uh perhaps
worth mentioning here is that uh fpp and
active inference uh is employed to
describe any uh complex system or any
self-organizing system
um that looks as if it takes uh the
action through which
um the free energy or that Norm I was
talking about is minimized or optimized
uh regardless of whether the agent of
Interest has any conscious awareness of
its intentions or not so it can be
applied both to
conscious cognitive agents as well as to
uh I don't know anything that's that
only has uh very simple basal cognition
and uh
very simple sense of
agency and
goal-directed behavior so that makes fep
an active inference broad enough to
Encompass all of these systems
um under a similar or even an identical
mathematical technology
yeah thank you Ali just like the linear
regression can be used across different
skills the system all the work is in the
modeling

or so
it's
this is like the textbook is
um oriented towards people on the first
half who want to learn about the
topic
also this is the first textbook so it's
kind of like
we bring a lot of other things that have
happened since and people can continue
to bring things in because it's just one
view but the first five chapters are
more on learning
active inferences then the second half
even though
we don't have all the kind of pieces in
place but making the generative model
the second half a lot of the very
philosophical very
general questions
it becomes easier to partition the
questions into like
what's a question more about an
implementation in a given system of
Interest
and then what is the question about the
more General
um usage of the term
but to kind of return all the way to the
initial point which is certainly felt by
all like
the active inference ontology which is
to say like the language and the terms
like is
some terms that are used uh commonly but
used in seemingly a different sense
here's one way that through the active
inference ontology we like at least

start to make sense of this and and
you're always welcome to use the at sign
to like connect it to an ontology term I
don't know if every single word but it
benefits to see it that way
um but like it helps emphasize that it
isn't an infinite slash open-ended set
of terms that are being used
um it's from a few
subfields where terms are being used
exactly and specifically and technically
mostly from Bayesian statistics
there's also some terms from on the
inbound and on the sense making side
from like signal processing ecological
psychology as well as on the outgoing
more action and control theory so
there's those kinds of terms
that describe like broadly cybernetic
agents are being used essentially
exactly
and then
for many of the terms these are just
proposed
just which may or may not be used
exactly By Any Given author but a lot of
the terms have like a narrower technical
sense and then a more General usage like
attention or anticipation or normativity
it's like it like can be used in
seemingly different or
um a narrower broader way
there's a lot to say in in how we all
I guess learn and uh
use the terms correctly another piece to
add just on the ontology is
um the terms are not just mostly drawn
from these computational areas

but
the the
I I always look at that anyone else want
to add something
oh just on that following that path uh
I'd like to clarify
a few things so when we're talking about
surprise
we're using that term
essentially in in the terms of like
entropy
in information Theory more or less like
the Shannon entropy
correct and slash if so when we're
talking about because minimizing
surprise turns out to be challenging for
technical reasons
you know we use variational free energy
as a proxy
um so I just like a little Clarity on
those technical reasons that minimizing
surprise is so difficult is that because
it's just too broad of a
series of possibilities for a Bayesian
calculation
um and then so what's the and then
what's the mechanism of using free
energy as the principle
if it's because we're just we're doing a
Divergence
that's my understanding
but clarification would be
nice
yeah Ali go for it
uh yeah we'll come to these questions
um uh in chapter two I guess uh and
particularly in uh on page 19 and uh 20.
uh so uh yeah the idea here is uh

although uh the value that needs to be
optimized exactly is that the free
energy but in practice uh optimizing or
minimizing this value
precisely or uh in its exact uh Bayesian
inference way uh it would prove
intractable so
the way to somehow solve this solve this
issue of intractability is to provide a
kind of lower bound that needs to be
optimized called variational free energy
so uh except for very simple scenarios
in every real-time uh where a real-time
a real real world situations and there
is this
um lower bound that needs to be
optimized and not exactly the free
energy or the surprise although the
theory of active inference uh evolves
around this idea of optimizing
um just a specific normal value namely
free energy but uh in practice uh that's
the variation of free energy that needs
to be
um optimize as the lower bound of
surprise so yeah a surprise or surprisal
is used and it's a statistical sense in
active inference literature and it
doesn't exactly map onto the
psychological sense of the term uh and
um again it's a technical word
and it doesn't necessarily refer to it
doesn't require any
cognitive or conscious agent for this
kind of informational or statistical
surprisal to be defined it's just uh
yeah something that can be defined
purely in its mathematical terms

using the parameters of the system of
um yeah inks
right
so we come from Linguistics so I plan to
learn all this math very soon but my
question is this
um the low road and the high road
are they like the same oh you guys can
see because I've blurred
are they like the same Journey from two
points of view like a tube which is a
microscope if you look on one side and a
telescope if you look on the other side
or are they completely traversing
different domains the way like syntax
and semantics they do meet and they kind
of like interlock but they are
they they walk the chess board with a
different Stitch step
Manuel thanks
um just before uh I guess addressing
that I wanted to
say how I understand the limitations so
I took to Jesse some I don't know if I'm
able am I able to share my screen so I
have a presentation I've been planning
to give this to my to my research group
so I have sort of like visuals to to
explain at least the way I understand
why minimizing free energy
it's it's a it's a good alternative so
if is it okay if I would sure how long
is the presentation no it's like three
slides just to share the concept of why
integrating like why Computing all right
surprisingly go for it very very fast
yeah go for it
so basically can you see my slides yep

Yeah so basically you know we're in the in the task of computing based ethereum so Computing some latent variable data given in our observation sex and this always involves an integral of this form so the denominator of of Base theorem uh can be computed basically integrating the numerator through all of the possible and unobserved uh uh or latent variables and the way you will do this computationally if you're if most of these integrals don't have a closed form solution so you cannot solve them analytically so what you do numerically is you would imagine you have like a grid and you evaluate so you don't know the height of this function and what the way you're going to integrate is by dividing the grid into different points and then at each point evaluating the height and the integral is going to be basically sort of like based and hot times height so you're going to add all of these small contributions but if you look at this example that I drew here most of the evaluations don't contribute at all to the to the integral because they're all at zero but you don't know that to begin with so evaluating this numerically will become incredibly challenging and furthermore the more the dimensionality of your problem uh increases like the more Dimensions you have the more variables you're trying to infer the more and more and more grid points you will have to evaluate in a so it grows exponentially so it just becomes in in

high dimensional spaces evaluating these
would become just basically impossible
numerically
so what we propose is to say okay what I
really want is my posterior distribution
latent variables given my observations
and I'm gonna propose a good enough
approximation
so this is my true belief that I want to
get and this is my proposed close enough
so what we'd say is like okay I want my
close enough approximation to be as
close as possible as a true model and
that is done by minimizing these scale
but you know that minimizing that
distance or pseudometric already
involves the distribution that I don't
want to work with the one that is really
hard to compute but then you do some
math that you'll see in the book I guess
and you arrive at this that the free
energy is uh a lower bound on what we
want to compute so basically I don't
know how to compute the right hand side
of this equation but I know that if I
make the left hand side of mass equation
as small as possible I would have also
made the other one as small as possible
given my approximation so so that's how
I understand it and there's different
yeah I guess that's that's good enough
for what I want to say just to shed some
light on the technicalities of actually
implementing these things
computationally uh it just you need a
proxy because there's too many
dimensions to handle
awesome cool

yeah very
um relevant it's discussed a lot in live
stream 52 series with Lance DeCosta some
of the relationships with like sampling
based methods variational methods all
these things
um and yeah
um Susan and then Rob Moore
so yeah that was very helpful
um it kind of speaks to the requisite
variety aspects I guess
um and and that kind of connects to
in terms of belief updating and learning
um yeah I know that there's
variation that is more in the moment
more emergent but the and that there's a
time lapse in the belief updating but
would
um would that mean
that the goal of learning and updating
is widening those boundaries
foreign widening
um yeah the the requisite choices
yeah there's a lot to say there it is
related to the cybernetics the principle
of requisite diversity and good
regulator theorem and those are
discussed a lot in the
chapter as it's kind of like
um like app and then uh often these
these three scales of inference
um for for so as Alia emphasized there's
like continuity with the simple or
simplest systems like things that don't
have these like higher cognitive
functionalities at all
um things that do only trivial belief
updating like even the case of a ball on

the ground you could think of it as
doing inference and doing nothing and
being at equilibrium
so it's a measurement framework that on
one hand you can have it totally
non-cognitive or basically cognitive
like ball rolling down a hill truly as
doing some path of least action that's
exactly from mechanical physics and the
Bayesian mechanics extends that ability
to model like mechanical systems and
then introduces the faster level of
inference and then attention which is
precision modulation
and then learning
which is the slower parameter updating
that's within one time within one scale
of of uh time and then nested models
could Implement arbitrary other learning
schemes
like here's just another a switch like
higher variable that just switches
between these knees so that's not
learning in the sense if those
parameters might be fixed
so it's like a narrower statistical
learning definition versus like broader
human learning
so again that's another term like
learning and inference and updating or
they're not being applied to the human
psychological
usage it like learning is used in
statistics to describe parameter updated
why I'm interrupting so but this model
does
um allow for human learning
and I guess would human learning be or

you know if it doesn't say so now but uh
um and are there model models around
policy
in terms of
you know if the
if the goal is human learning
I didn't want to speak to that
all right go for it
so
um I don't have an answer to your
question Susan so I'm going to
uh ask another question and maybe if
someone has a comment
uh for you they can think about it in
the meantime
uh my question is related to something
that's called Kristen touches on a bit
in the preface
and then it's it's discussed
um another
summaries of the free energy principle
that you could find for example
Wikipedia
and it's the
uh falsifiability
of the free energy principle and
a distinction
that you might make between a framework
or in a principle versus
a a theory that's making
testable or uh
measurable
predictions
uh that would allow you to confirm or
reject
a
provide evidence for or reject a theory
and

I would call Kristen and and Thomas Parr
balutu in the book is that the free
energy principle is is just that it's a
principle akin to the theory of
evolution
and and somehow I haven't been able to
make uh sense of what what it is about
the free energy principle or or the
theory of evolution
as a more
a common uh Theory or framework that
people are more familiar with what is it
about those
the principles or Frameworks that makes
them hard to
falsify or not classifiable
and then they go on to specify that when
these Frameworks do become testable is
is when you introduce the generative
model it's an active inference the core
of
the of the Theory comes to to
producing a generative model producing
some description of
what
um causes
uh in in the worlds generates
uh could could be generating the
observation is that you're you're seeing
and and then that's something that you
can go and measure maybe in a human
brain or or look for
observations to to test against but uh
so the essence of my question is what uh
distinguishes
the free energy Principle as a framework
that isn't
falsifiable

and how does a generative model
converted the free energy principle or
active inferences to a theory that can
be tested
great question Ali go for it
uh so the reason free energy principle
is described as
um as an unpalsifiable principle is
because it's basically a variational
variational principle uh which we use to
we use as a mathematical Tool uh to
describe the kind of theory
[Music]
accounts for the behavior of uh
particular kinds of self-organizing
systems or agents but
if we look at this issue in a much
broader perspective uh the same problem
of falsifiability applies equally to
predictive coding Frameworks as well so
the question around is predictive coding
theories are for false the Bible or not
has been going around
for several decades now but a very
Illuminating uh paper published just a
couple of days ago namely is predictive
coding falsifiable by Brown by Bowman at
all in which they argue that although it
looks as if these kinds of theories as
predictive coding or active inference
and so on looks as if they are
falsifiable but the fall
survival sorry but the the
unfalcifiability of these theories is
nothing inherent to these theories at
all uh it's just the problem of
um Precision
that needs to be accounted for if we

want to look at the problem of
falsifiability for these theories so
their conclusion is that these this kind
of theories is predictive coding or
predictive processing and active
inference can indeed be viewed as
falsifiable Theory at least in theory
so it's nothing
not there isn't any inherent barrier for
describing these theories as
unfalcifiable there isn't any
fundamental on falsifiability for these
theories but then again when we uh if we
come back to active inference again
there are two parts for
um for the theory of active inference
the underlying variational principle
namely pre-energy principle and then the
empirical Theory cell namely the active
inference so even if the underlying
principle or fep is unfalsifiable it
doesn't necessarily apply to active
inference as well so uh active inference
can be viewed as a perfectly empirical
testable and falsifiable Theory
but there are some arguments that the
underlying principle uh being as
variation for a principle is in essence
on falsifiable so
yeah
thank you ollie there's a lot there um
you can always follow where I am in
Dakota by clicking on the icon to like
improve the notes which would be really
helpful or add more information but in
this interview
that was where friston provided this
response

that's one answer to this kind of
question another is like again to
connect it to the linear regression like
the statements about variational free
energy bounding surprise minimization
and then surprise minimization being
equivalent to model evidence
maximization
they are two sides literally of the
exact same thing if you have the optimal
model
you have the lowest bounded surprise
lowest surprise that's the exact
Bayesian results
it's like you're you're doing the best
that you possibly could have on the
inbound
sense making side and then on the
outbound control site so again a
deflationary statistical
description which is accurate of
variational free energy is it being a
unified
tractable bound to surprise minimization
whereas there isn't attractable
analogous bound directly to model
evidence maximization
which is why there's the whole question
about
hill climbing algorithms
and and model maximization but this
understanding between minimizing
something is like equivalent to
maximizing something is commonly used
like minimizing the training loss on a
neural network or minimizing the L2 Norm
like the sum of squares on a linear
regression

it as was mentioned by someone earlier
like that doesn't mean that you made the
generative model
adequate to the real world in any way
it's just a statement that the
statistical model
will have a certain
um ability just to process the data
routinely like a t-test
does that mean that that was an ideology
for a t-test or something that that has
always been the question
that is a separate question than the
relationship between variational free
energy and surprise which is just a
technical
finding
that it can't be falsified it's just a
construction made
but then once something is proposed like
as a structural hypothesis like in the
SPM days they were testing does uh
presenting a then B influence this
response pattern and then testing
structural hypotheses in the brain
active inference kind of elaborates on
this idea of creating structural
hypotheses
that themselves being part specific or
particular
and specified
or just evidently not
it'd be analogous to asking about like
an unspecified linear regression whether
you'd expect this or that result
let's look at a few other of the
questions
can I quickly make a comment about uh

Russ

a question I recently read this paper

the math is not the territory navigating

the energy principle but I um mayor

Andrews and you know my takeaway from

that paper which I highly recommend

it's that the way to think about the 300

principle is more like uh statistic yeah

like a statistical

uh framework like you don't falsify

calculus calculus is uh you know

mathematical framework but if you apply

Newton's laws and let's say you have an

alternative hypothesis to Newton's

clause and they don't agree with the

data then those laws are wrong but

calculus is still right like calculus is

still the mathematical framework to

analyze you know differential equations

so in that sense uh the 300 principle I

think sort of place an analogous role

where now the falsifiable thing will

cause us then I was saying like you

build your generative model maybe you

think about Neuroscience you think about

how the brain connects or maybe you

think about I don't know biophysics how

genes talk to each other so in that that

is what is going to be falsifiable on

the on the free energy principle

it's just a way to look at that genitive

uh in a in a way to make sense of it

like uh what you're proposing kind of

thing so so that's my understanding

might take away message from from this

paper uh yeah

yeah great can you please uh link the

paper

in the yeah live stream 14
also it was discussed with melon with
others but yeah that's kind of like a
key reference point to help clarify
some of this like map territory and the
map territory fallacy and the map
territory fallacy fallacy we discussed
this actually earlier in the cohort for
discussion if you wanted to look but
let's look at our last like little bit
to some of these written questions
while the fpp is applicable across many
time scales
how applicable is active inference as a
process theory instantiation of the fep
very much plays into our discussion do
organisms need a nervous system for it
to uh make sense to describe them as
doing active inference no
you don't it can include inert systems
passive systems
this isn't the psychological account
but it can be applied to
mind as a system of Interest
but the bazel Machinery that is
described in this textbook
um like what chapter 4 describes
generative models of
are not psychological accounts of humans
their statistical examples
so now there's no need for any kind of
specific material
or to have to to be material at all it
could say oh it's a matrix that just
flips between two states
it could be as simple as that or you
could be applying it just like another
analytical or measurement or modeling

tool
to a system of Interest
so and how
how applicable is active inference
that's the empirical question
I'm hoping to get intuitive
understanding of how free energy
represents a measure of the discrepancy
between the organism's internal model of
the world and its sensory inputs what is
the underlying relationship between
information and energy
what is the relationship between this
free energy and active inference and the
thermodynamic concept of free energy
that's also present in any organism
environment
anyone have a thought on this what's the
relationship between
variational free energy or expected free
energy
the thermodynamic concept of free energy
and I say share my thoughts on that yeah
go for it
yeah so
my understanding is the following so you
know in the industrial revolution people
are trying to improve uh steam engines
and they counted the realization that
doesn't matter you know how good your
design is how well you know that as
perfect as you as your aging because
you put some energy but you don't get
the the same amount of work out of it
there's always some loss there's always
something that you cannot really utilize
so that leads to Klaus used to make the
statement of like oh let's just come up

with a concept that we're going to call
entropy which is always increasing and
it's kind of like we this part of of the
of the energy that I am putting to my
system that I wasn't able to to make use
of like it's just dissipating you can
think of friction on other things that
just make the the
the the all of the energy that you put
into the system not you not been
utilized
later on boltzmann comes in and then he
says look entropy is not mysterious
entropy is just a measure of how many
ways you can arrange a system how many
different you know if I have the gas
molecules in in this room uh nothing in
the loss of physics uh prohibited them
to certainly like you know have some
random trajectories and collecting that
corner and then I'll be suffocating
because I'm in a vacuum kind of nothing
in the laws of physics for bits that but
it's just incredibly unlikely it's just
there's so many many many more ways to
have them uniformly spread throughout
the room that you'll have to wait longer
than the age of the universe to see that
random event of having them in that
corner
so now when you uh Shannon comes in and
he
um formalizes the idea of of uh
information as a mathematical quantity
that you can
put a number on turns out that the
metric that he uh derives is exactly of
the same functional form as Boltzmann's

entropy so you know the two equations
look exactly the same and The Story Goes
that I think Von Neumann told Shannon oh
just use the word entropy that is the
same equation and nobody understands the
other one either so an advantage you'll
win
so then later on it I think is EP Jane
So a very famous and important physicist
in all of these uh statistical mechanics
and its uh relationship with based in
statistics that I think finally makes a
connection between thermodynamic and and
informational entropies in that case
where uh from my readings and
interpretations of what Jane said is
that thermodynamic entropy is something
I can measure I measure the amount of
heat dissipated by my system at a
constant temperature and I put a number
that's my entropy
the information entropy is not something
I measure like that it's it's in units
of bits but it has to do with
in a system if I push a piston
how much information I or how many
degrees of freedom I control of that
system well I can push all of the atoms
in the gas and compress them but I don't
really have control over their
particular trajectories so the work that
I get out of that effort has to do with
how many uh degrees of freedom I
actually control if I was able to go
into the atomic level and point all of
the gas molecules into the same
direction my piston will be much
stronger and like push and give me more

work but I don't have access to those
degrees of freedom so the information I
have access to uh is limited therefore
if I were to reproduce that piston
pushing and uh one or over and over
again
it will give me the same result given my
limited information
so I don't know if that's helpful you
know it's uh it's uh it's a long uh
explanation but uh that's my current
understanding of how these two things
are connected
thanks yeah
Kristoff go for it I just want to say
something present this from a slightly
different angle this is something that
um I I took away from my undergrad um I
particularly studied uh uh two kinds of
semantics of programming languages
denotational semantics and operational
semantics which is you know you take a
program fragment and you're trying to
assign a meaning to it and the
operational semantics is some kind of
sequence of reductions using big or
small steps that tells you like you know
how the program evolves over time and
what the sort of State transitions of
the of the virtual machine kind of are
and there's a different way of assigning
semantic to to a program is to say well
there is a mathematical object
um that this program represents a good
example that would be for instance
something like like an infinite list
it's not actually realizable in in a
finite computer and yet we have programs

that act as if they were in infinite
lists so we're really hit me is
specifically in this in this case was
that um you know if something has the
common presentation of the of the issue
of entropy in uh information Theory
versus thermodynamics is that people
often mentions like oh these two things
have the same mathematical form but
there are two different things
and I would say that because these
things have the same mathematical form
there must be the same thing they must
be the same thing
um it's just the the the depth and the
nature of the relationship may not be
precisely understood by us just yet but
if the mathematics suggests that this
it's precisely exactly the same
phenomenon because it has the same
mathematical form
this is kind of a more philosophical uh
approach
thank you Kristoff
another angle so that's a philosophical
kind of uh
angle uh
one is to say A variation for energy is
an information theoretic free energy
that has that resonance and form or the
analogy and form and then when we apply
this to self-organizing systems then
they if they if we look at things that
are existing already
they are representing a small
um like they're unlikely in one way but
then they're likely in their own way so
perhaps one way to see it is it's about

finding what makes it have a thingness
which is what Ali alludes to in the chat
with uh fpp and a theory of everything
again thing here is used in Iraq that
are specific ontological sense
definitely we can return to this like
well and it's not many times with like
this notion of particular States and a
particular thing
more to say but um Susan
um yeah that was that was a great
explanation manual and what piques my
interest and I just want to put a pin in
uh what you talked about was um the um
the correlation between that and the
constructal law
and yeah I think that that's
you know as far as
how I kind of tease that apart is is
is um
the confusion over who designs and I
think
you know that uh that has a bearing in
you know in our updating belief updating
so that'd be interesting to pursue I
don't know that it's
covered in this course but it would be
an interesting
exercise outside of it
cool yeah cool guys quickly Point people
if um there's a great great paper
um what is information
by let me find in my uh David do you
remember the author
it's just that's I think uh
if I if I find it I'll paste it on the
on the chat but that just explains for
these ideas of entropy you know

entropies in the eyes of the beholder
like the entropy of a coin is not one
bit if I only fire Focus only on the
faces yeah it's one bit I just can't
have either heads or tails just one yes
no question but what if not I not only
care about the face of the coin but the
angle when I toss it like this is like
heading this way or this way and I can
discretize it into four corners or eight
corners or as many as I want
so the question of what is the entropy
of the system it's really a matter of
who's asking the question and what they
want to answer in a sense uh
so I thought that is a great point and
the relationality
is a theme that comes up again and again
like just the fundamental octave role of
the interactance
like the passive Observer and the kind
of receptive filter in only like all
these ways that the active inference
helps reformulate that relationality
like even in the colleges that you're
mentioning and then the strongest
statement I believe
um to I mean to christoph's point
uh of going further like like the
Overton window on this side I think is
Alex Kiefer in some of these discussions
where he's like it is for like it is
further that it is informational but but
the the um the actuality of biological
systems makes it so that this
theoretical construct
has like
a stronger than

um instrumental
relationship with thermodynamic
free energy like if you had the best
foraging algorithm your colony is going
to exist in the future the best foraging
algorithm doesn't mean the most it just
you know means given the niche
so that's a normativity that has like a
survival normativity to things that
survive
so that's another piece of how it
relates to today Migos
yeah it sounds like there might be some
unifying concept here that
both in the
rmodynamic concept there's an energy
gradient that has some potential utility
for some agent with some goal and in the
information there's also from some
specific perspective
the potential for harnessing that
there's
if that makes sense it's like if you
take a specific vantage point of an
agent within that system you can harness
information or energy gradients to some
end result
yeah that's very resonant with Chris
fields and Mike Levin's work on like
poly Computing which is that the
computational
attributes are themselves relational
so like if something is doing a totally
encrypted computation it can't be
distinguished from noise
but that's like kind of like a um but
then encryption is something that is
procedural whereas so that has a

different
set of formalisms than like a
statistical distribution
the the those are all really
good points and ones that that that
connect very provocatively but not
you know directly necessarily to
some of these Felts ways of thinking
about that which
um yeah
interesting times and discussions does
anyone want to like add any other
comments in the last minutes
well hope that was a fun first
chapter discussion
we'll have the earlier time next week
people can add more questions
it add and and update the the answers
that they're finding relevant
um often we have like yeah more people's
General Reflections in the first of the
discussions and then try to address any
and all that because that makes us think
about so many things after these
discussions that people can just add
them here
and uh yeah any other comments or
thoughts or we'll end it
I just want to say there's some very
enlightening discussion towards the end
and uh thank you all for the
contributions
yeah thanks a lot guys thank you
everyone thank you
bye thank you Daniel